
Ransomware Monitoring Script

This Python script (also executable tool) can monitor a specific directory that you can choose and look for signs of ransomware: either multiple and rapid changes in filenames or suspicious file extensions or even some suspicious process names.

Main functions:

- **Process Monitoring:**
 - Detects running processes names that match a defined list of suspicious or ransomware names. This list can be customized.
 - Alerts the user if a match is found.
- **File System Monitoring:**
 - Monitors directories and detects rapid file modifications, creations, and renames.
 - Monitors renames with suspicious file extensions like .locked for example
- **Customizable Parameters:**
 - Can configure the directories to monitor and the list of suspicious process names and file extensions.
 - We can modify the maximum file modifications limit and the time to reduce false positives.

Requirements

- Python 3.6+
- Required Python libraries:
 - watchdog
 - psutil

Install the dependencies using pip:

```
pip install watchdog psutil
```

Usage

1. **Edit the Script:**

- Open ransomware_monitor.py in a text editor.
- Customize the following:
 - monitored_dirs: Add directories you want to monitor (example, "C:\\Users\\Testransomware\\Documents").(use \\ instead of \ because of escape characters.)
 - SUSPICIOUS_PROCESS_NAMES: You can add any process name you want.
 - SUSPICIOUS_FILE_EXTENSIONS: You can add any extension you want to look for.

2. Run the Script:

Python3 ransomware_monitor.py or just click on the executable.

3. Monitor the Output:

- The script will log warnings to the console for suspicious activities.
- Press Ctrl+C to stop the script.

Example Configuration

- Monitor the user's Downloads folder:

```
monitored_dirs = ["C:\\Users\\<username>\\Downloads"]
```

- Add suspicious process names:

```
SUSPICIOUS_PROCESS_NAMES = [ "ransomware.exe", "cryptolocker.exe",  
"wannacry.exe", "locky.exe"]
```

- Add suspicious file extensions:

```
SUSPICIOUS_FILE_EXTENSIONS = ['.locked', '.encrypted', '.ciphered']
```